Asset Demand of U.S. Households

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Summary

- Monthly data on portfolio positions and flows of over 200,000 US households including Ultra High Net Worth Individuals (UHNW)
    1. Security level (Apple, Google etc.)
    2. Narrow asset class (US Equities, Corporate Bonds, MMFs, Munis etc.)
    3. Broad asset class (Equities, Fixed Income, Cash, Alternatives)

- Provide a host of summary statistics on household rebalancing across asset classes: Many new moments to inform macro-finance models
    1. Low flow-return sensitivity of households
    2. Less wealthy households act pro-cyclically, UHNW are contrarian
    3. 3 factors explain 75% of HH rebalancing behavior across asset classes

Comments

Great paper!

- Really comprehensive summary of massive dataset
- Provides a more complete picture of inelastic financial markets: Households do not undo institutional inelasticity
- Link between household finance and macro-finance

This Discussion

1. Cross-Section versus Time-Series
2. Who absorbs financial fluctuations?
3. What drives the heterogeneity across wealth groups?

Cross-Section versus Time-Series

- Calvet, Campbell & Sodini (QJE, 2009) define active change of household h as the chg in risky weight accounting for price change

$$a_{t,h} = \log w_{t,h} - \log w_{t,h}^{Passive} \approx f_{t,h}^{Eq}$$

$$p_{t,h} = \log w_{t,h}^{Passive} - \log w_{t-1,h} \approx r_{t,h}^{Eq} - r_{t,h}^{Portfolio}$$

Where $f_{t,h}^{Eq}$ is the flow to equities relative to equity wealth (≈ Liq Wealth)

- They then run the following cross-sectional regression:

$$a_t = \gamma_{0,t} + \gamma_1 p_{t,h} + ... + \epsilon_{t,h} \quad \rightarrow \hat{\gamma}_1 \approx -0.5$$

- GKMOY run a similar regression (averaged across HH):

$$f_t^{Eq} = \alpha_0 + \alpha_1 r_t^{Eq} + \epsilon_t \quad \rightarrow \hat{\alpha}_1 \approx -0.02$$

Time-fixed effects drive the difference in the rebalancing!

Households (partly) offset the idiosyncratic part of their equity return in the <u>cross-section</u> but not the aggregate equity return in the <u>time-series</u>.

Can households absorb financial fluctuations?

1. **Stock-level Elasticity** (Koijen and Yogo, 2019):
   Infer household equity portfolio as residual from shares outstanding and institutional holdings.

   $$\log w_n^{HH} = \alpha + \beta me_n + \epsilon_n \quad \rightarrow 1 - \widehat{\beta} \approx -0.4$$

   Households are the most elastic investor group! They play important role in absorbing financial fluctuations <u>relative</u> to other investors

2. **Asset-Class Sensitivity** (this paper)
   Regress household flows into equities onto equity returns.

   $$f_t^{Eq} = \alpha_0 + \alpha_1 r_t^{US,Eq} + \epsilon_t \quad \rightarrow \widehat{\alpha}_1 \approx -0.02$$

   Households are not very sensitive to equity returns! They play *"limited role in absorbing financial fluctuations"*, especially <u>relative</u> to institutions with $\widehat{\alpha}_1 \approx -0.2$ (see GK, 2021).

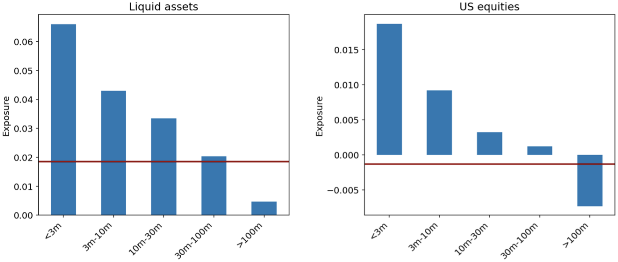
Can households absorb financial fluctuations?

- Aggregate asset classes less substitutable than individual stocks
  $\rightarrow$ Demand for asset classes less sensitive to returns than demand for individual stocks ✓
- But is the role of households relative to institutions fundamentally different when it comes to variation across stocks (1) versus across asset classes (2)?
- If you infer household holdings from difference between shares outstanding and institutional holdings (as in KY, 2019):
  Are the cross-sectional tilts correlated to the Addepar equity portfolio?

What drives the heterogeneity across wealth groups?

- Regress avg flow for each wealth group g onto US stock mkt. return

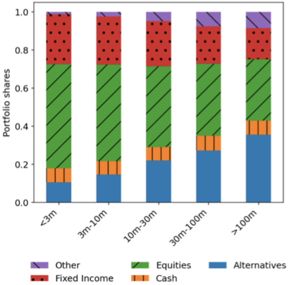
$$f_{gt}^{Liq/Eq} = \alpha_g + \beta_g r_t^{US,Eq} + \epsilon_{gt}$$



- **Very impressive figure!** What drives this result?
  1. Hedging demand & portfolio rebalancing
  2. Advisor Effects, Sophistication
  3. Endogeneity & Equilibrium

Hedging and Portfolio Rebalancing

- If households want target equity share, then rebalancing flows driven by **i) equity share** and **ii) correlation of equity with other assets**
- Wealthy HH hold lower share in equities and larger share in alternatives:



- If returns on alternatives and equity negatively correlated, wealthy HH need to act **more contrarian** to hold constant portfolio shares!

Advisor Effects & Sophistication

- Strong correlation between wealth and advisor choice. Do the patterns hold controlling for advisor effects?
- Intriguing finding: Excess volatility of flow to cash. Could be used as an instrument for wealth orthogonal to advisor choice?
- CCS (2009): Educated investors tend to rebalance more
- Are less wealthy investors performance-chasing/extrapolating at a monthly frequency m?

$$f_{h,m}^{Eq} = \gamma_{0,h} + \gamma_{1,h} r_{h,m-1}^{Eq} + \epsilon_{h,m}$$

Endogeneity & Equilibrium

→ *"Household sector plays a limited role to absorb financial fluctuations"*

- Ultimately all assets are (at least indirectly) held by households and hence driven by their preferences. What financial fluctuations do the authors have in mind?
  1. Idiosyncratic demand shocks by asset managers ($\neq$ households' preferences) cause financial fluctuations
  2. Not all households are captured? Is there a difference between households' direct and indirect holdings?
- Risk sharing in equilibrium:
  1. Bad news about the economy:
     → Risk increases → all *HH* want to hold less equity → prices drop
  2. In response to risk, less wealthy (more risk averse) households decrease equity share by more and UHNW by less
  3. Low sensitivity of flows and "elasticity provision" by the wealthy is just the side effect of i) aggregate shocks and ii) risk sharing in equilibrium

Conclusion

Super awesome paper

- Comprehensive, well-written summary of HH rebalancing behavior
- Host of (opinion-free) moments to inform macro-finance theory
- One step towards going full circle: Intermediary asset pricing + demand-based asset pricing + Consumption-based asset pricing

Comments

- Household rebalancing in cross-section versus time-series
- Maybe some verbal interpretation on equilibrium, financial fluctuations, and who is absorbing whose demand shocks?

Appendix

Summary of moments to inform Macro-finance models

1. Portfolio Choice
   - Weight in cash is stable across the wealth distribution
   - Wealthy households hold lower share in equities & higher share in alternatives
   - Weight in municipal bonds increases with wealth
   - Wealthier investors are more home-biased in equity allocation
   - Wealth does not explain differences in asset class weights ($R^2 < 5\%$)
   - ...
2. Flows/Portfolio Rebalancing
   - Household turnover increases in downturns
   - Wealthy households are less inert
   - Flow to cash excessively volatile relative to portfolio share
   - Avg flow to liquid assets and stock returns highly correlated (71%)
   - Flow disagreement increases in downturns
   - Household flows are inelastic
   - Flows positively correlated with returns for most wealth groups, contrarian for UHNW
   - Factor structure in rebalancing flows
   - ...